

Autoencoder, Diffusion model, contrastive learning and Medical adoption in AD

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1. Autoencoder

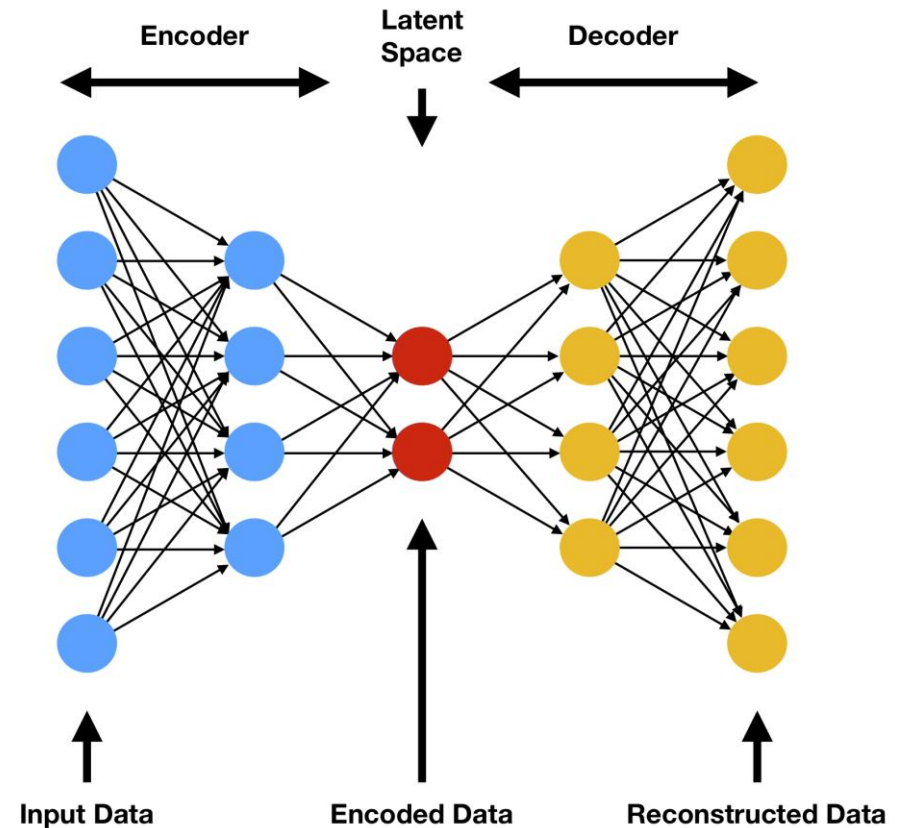
1. What is Autoencoder

Autoencoder란?

- 입력과 출력이 같은 구조를 가지는 network

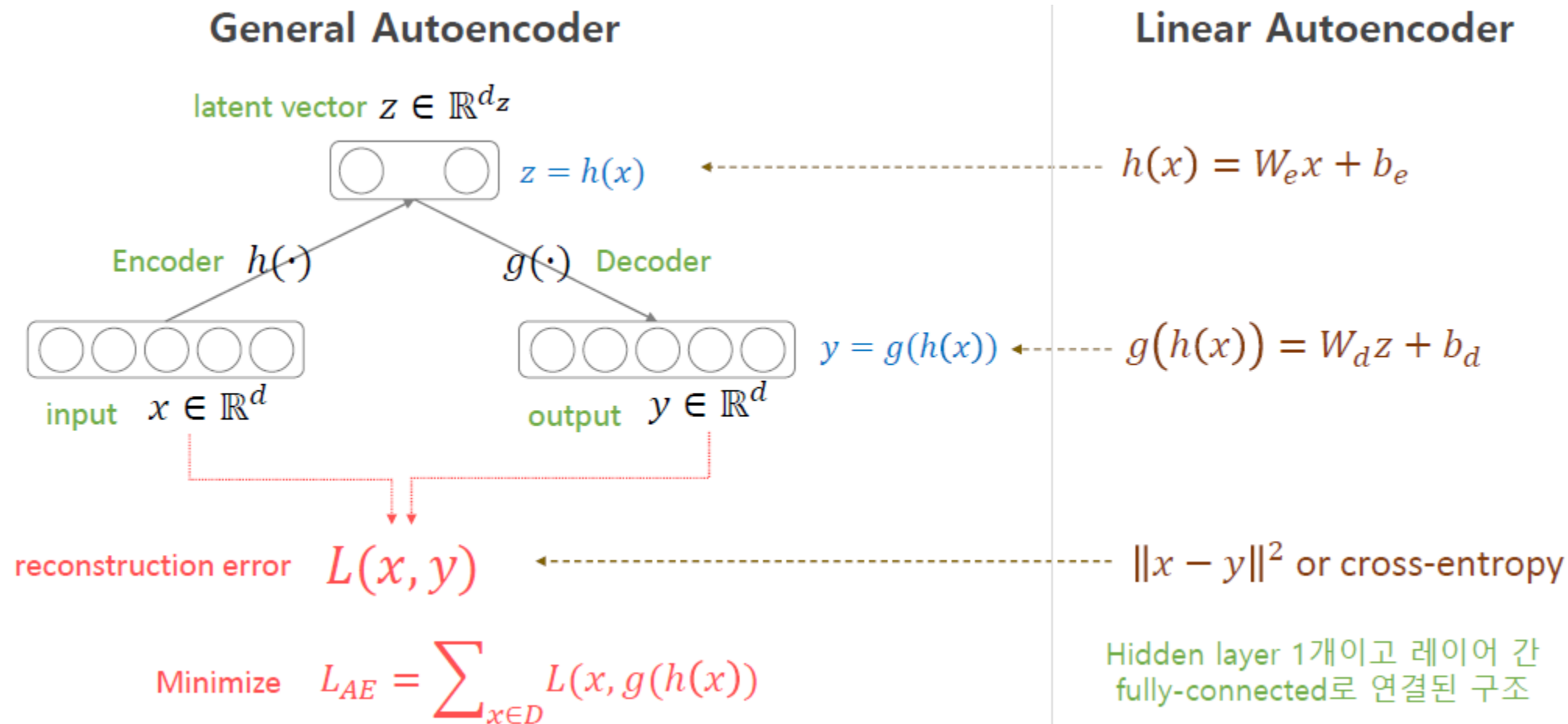
Autoencoder의 기능

- Unsupervised Learning
- Representation Learning
- Dimensionality Reduction
- Generative Model Learning



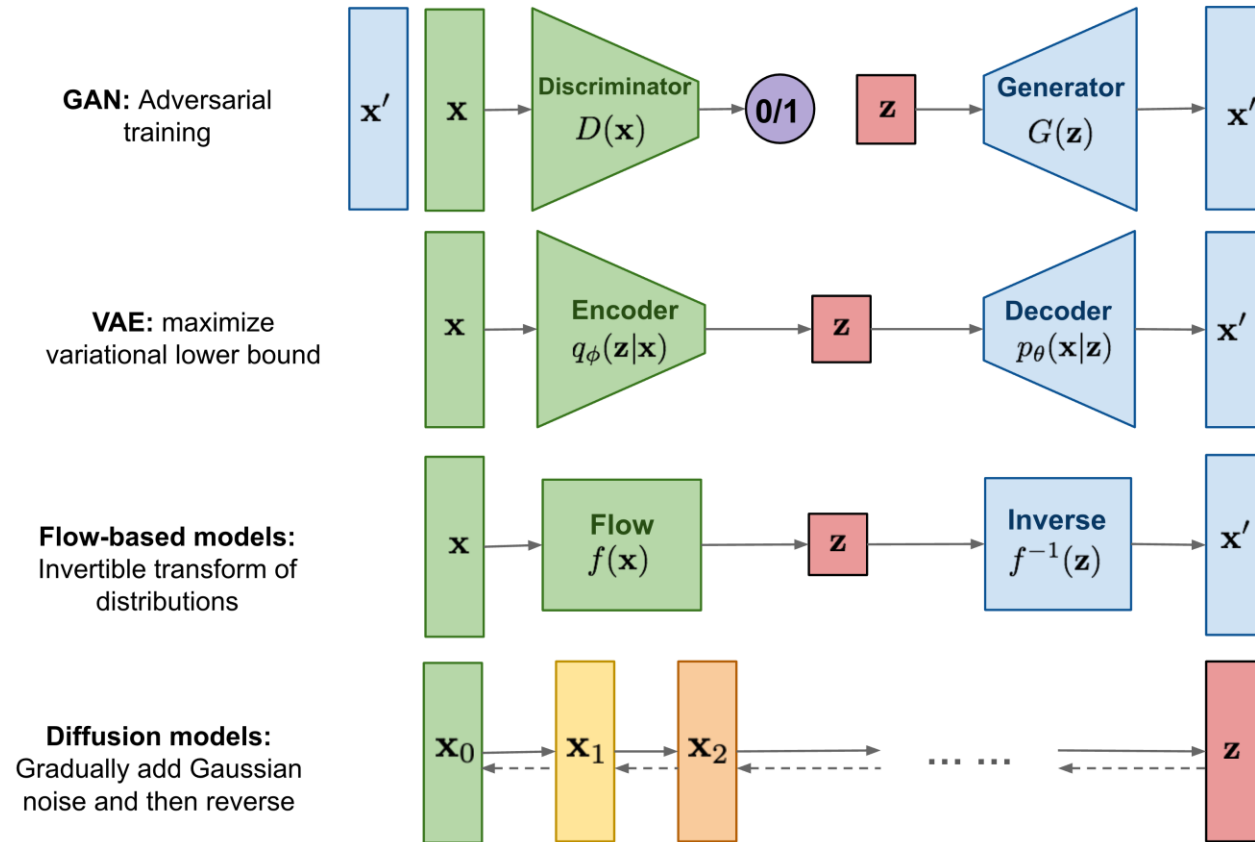
1. Autoencoder

2. Overview of Autoencoder



2. Diffusion Autoencoder

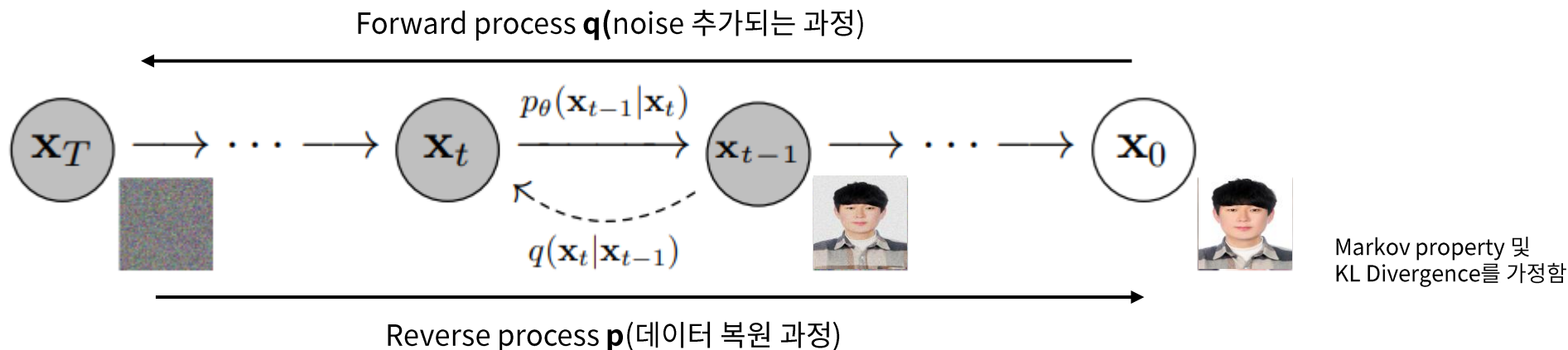
1. Generative model이란?



2. Diffusion Autoencoder

1. Diffusion Model이란?

데이터를 만들어내는 deep generative model 방법론 중 하나



Objective : 실제 data의 분포인 $p(\mathbf{x}_0)$ 를 최대화 하는 것을 목적으로 학습이 진행됨(Maximize likelihood)

Loss function :
$$\mathbb{E}_q \left[\underbrace{D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))}_{L_T} + \sum_{t>1} \underbrace{D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}} - \underbrace{\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0} \right]$$

L_T : p 가 generate하는 noise $\mathbf{x}_T$ 와 q 가 $\mathbf{x}_0$ 라는 데이터가 주어졌을 때 generate하는 noise $\mathbf{x}_T$ 간의 분포 차이

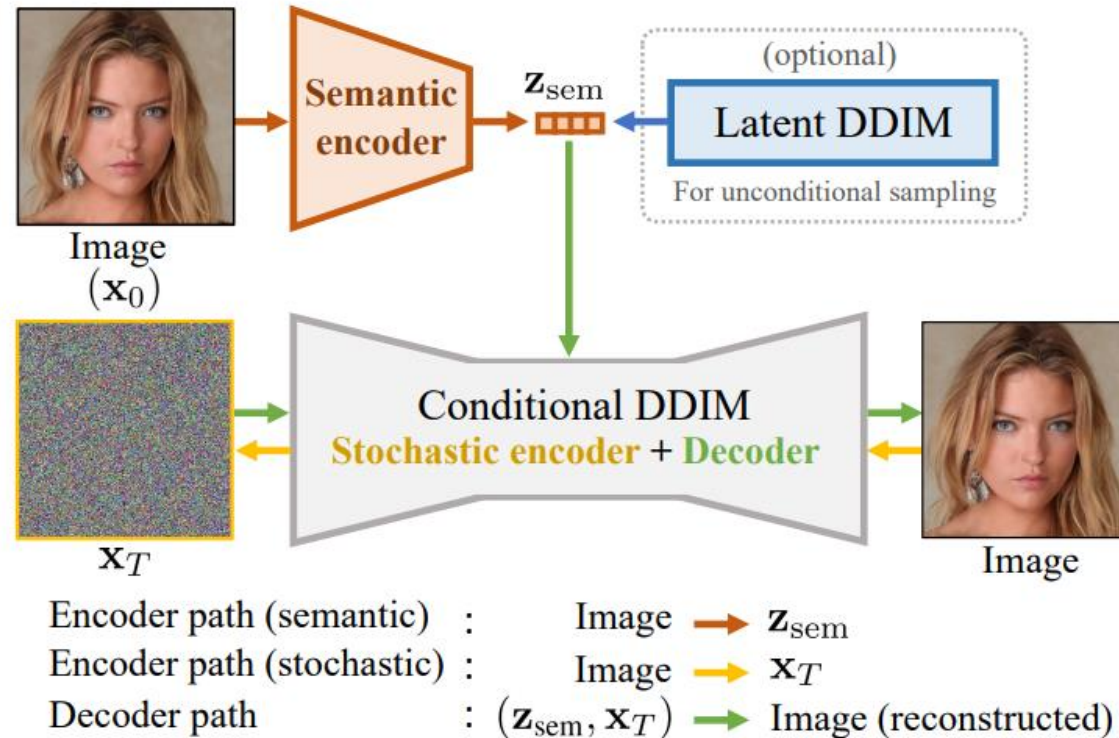
L_{t-1} : p 와 q 의 reverse / forward process의 분포 차이. 최대한 같아지는 방향으로 학습됨

L_0 (reconstruction loss) : latent $\mathbf{x}_1$ 으로부터 data $\mathbf{x}_0$ 을 추정하는 likelihood. Maximize하는 방향으로 학습됨

2. Diffusion Autoencoder

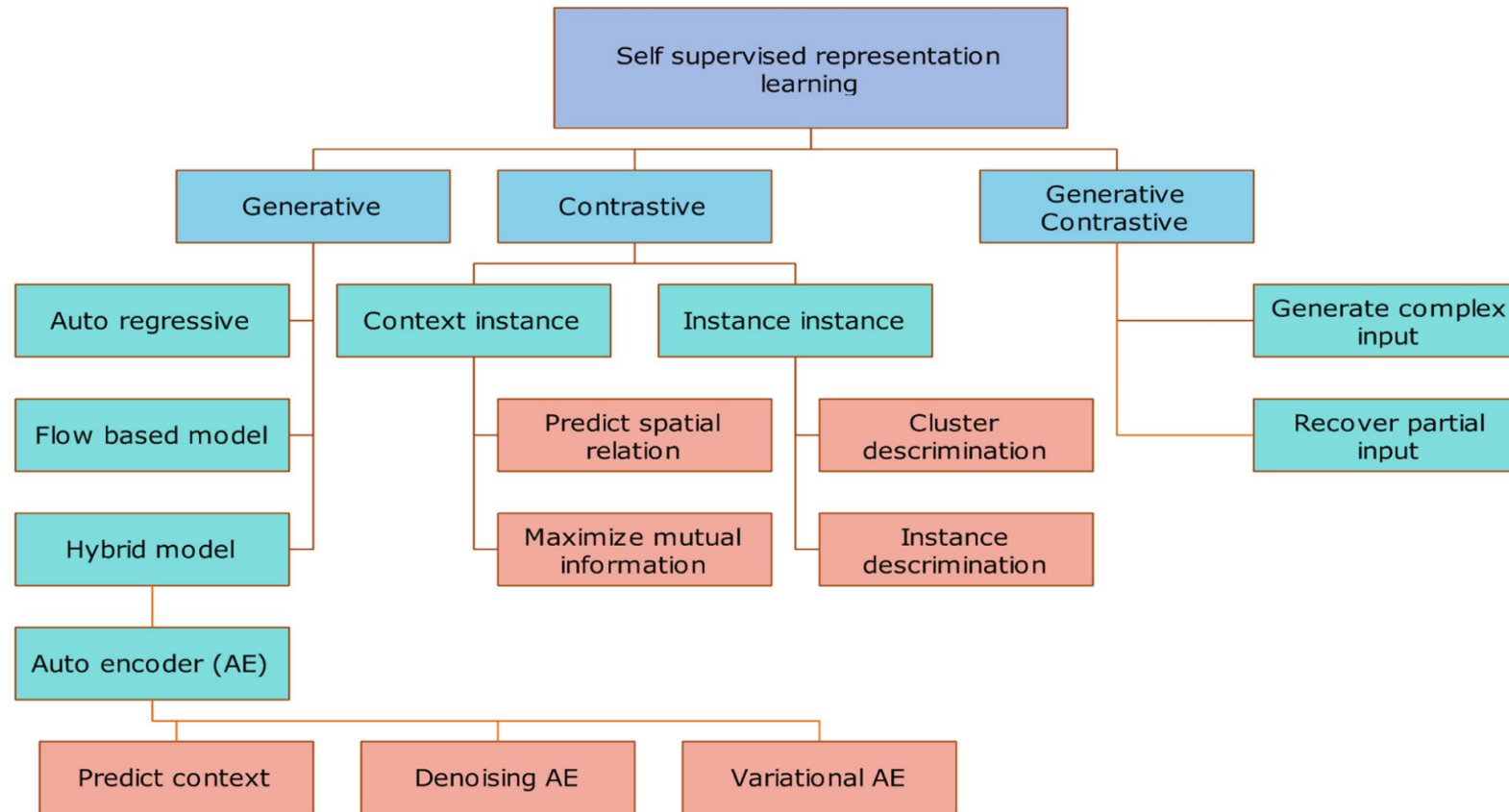
2. Diffusion Autoencoder

DDIM : DDPM이라는 Diffusion model의 한계인 semantic information을 학습할 수 있게 변환한 모델



3. Contrastive Learning

1. Self supervised representation learning

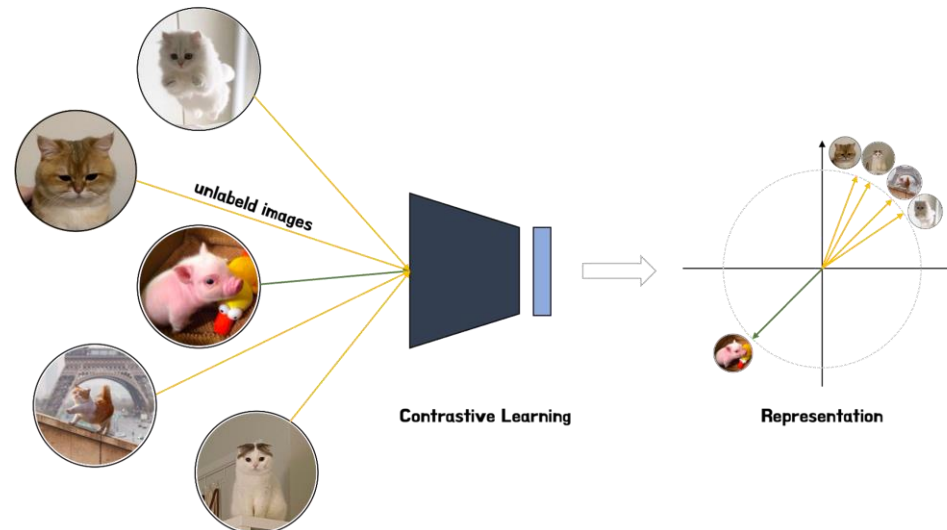


3. Contrastive Learning

2. Contrastive Learning

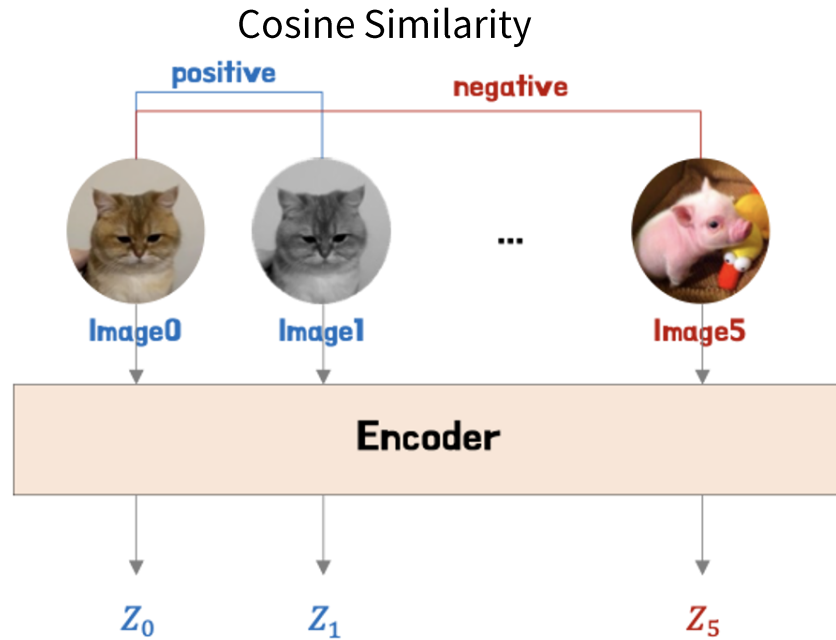
Contrastive learning이란?

- Self-supervised-learning representation learning 중에서 최근 가장 효율적인 방식
- 유사한 이미지가 저 차원 공간에서 서로 가깝게, 동시에 다른 이미지는 서로 멀리 떨어져 있도록 저차원 공간에서 이미지를 인코딩하는 방법을 모델이 학습하게 하는 방법론
- 기존의 생성모델 방법론들과는 다르게 특정 예측을 생성하기 위해 모델을 최적화하는 것이 아닌 표현 자체가 의미가 있는지 확인하는 작업을 거침 -> 레이블이 없기 때문에 이러한 방식을 채택함
- 따라서 주석이나 레이블 없이도 데이터에 대해 많은 것을 학습하고 모델을 훈련하는 것이 가능함



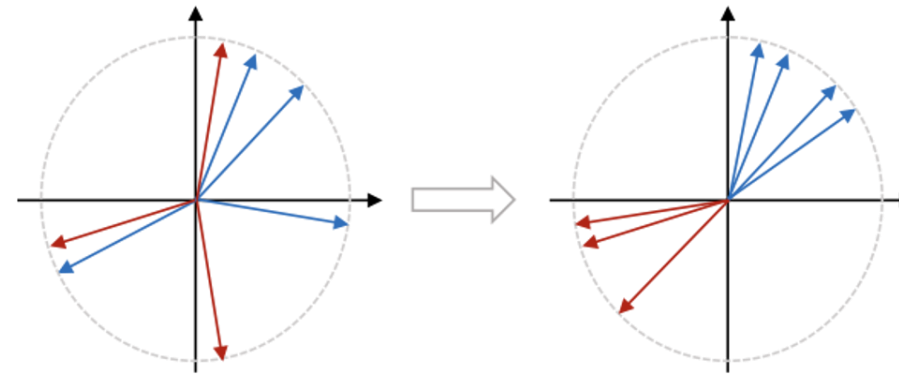
3. Contrastive Learning

2. Contrastive Learning



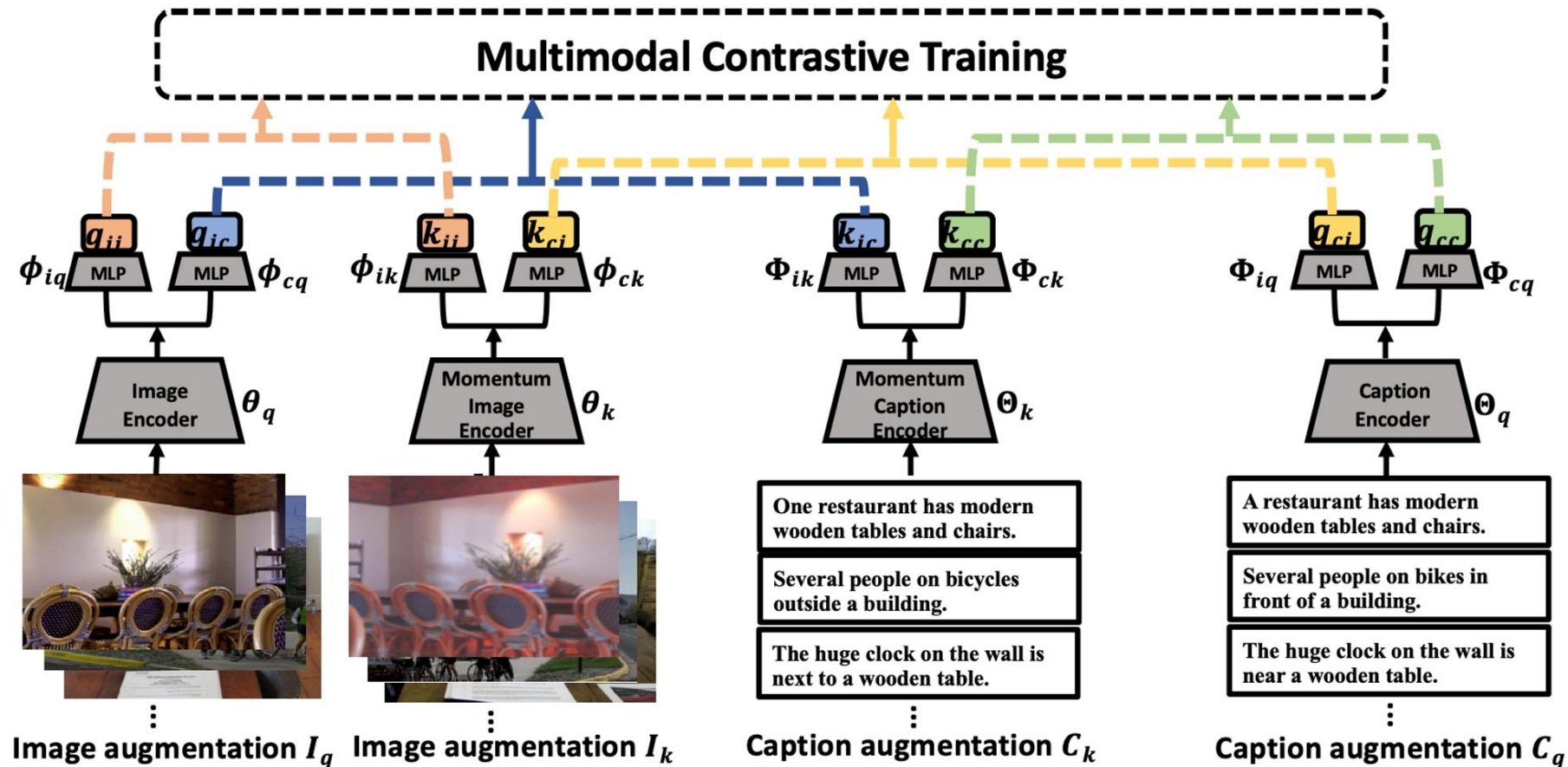
$$L_{contrast} = -\log \frac{\exp(z_0 \cdot z_1)}{\sum_{i=1}^K \exp(z_0 \cdot z_i)}$$

optimizing $L_{contrast}$



3. Contrastive Learning

2. Contrastive Learning – Multimodal Representation learning



4. Paper Review

1. Introduction

Interpretable Alzheimer's Disease Classification Via a Contrastive Diffusion Autoencoder

Ayodeji Ijshakin¹ Ahmed Abdulaal¹ Adamos Hadjivasiliou¹ Sophie Martin¹ James Cole¹

Abstract

In visual object classification, humans often justify their choices by comparing objects to prototypical examples within that class. We may therefore increase the interpretability of deep learning models by imbuing them with a similar style of reasoning. In this work, we apply this principle by classifying Alzheimer's Disease based on the similarity of images to training examples within the latent space. We use a contrastive loss combined with a diffusion autoencoder backbone, to produce a semantically meaningful latent space, such that neighbouring latents have similar image-level features. We achieve a classification accuracy comparable to black box approaches on a dataset of 2D MRI images, whilst producing human interpretable model explanations. Therefore, this work stands as a contribution to the pertinent development of accurate and interpretable deep learning within medical imaging.

1. Introduction

Deep learning models are increasingly used within the medical domain, which is one of the most safety critical fields (Vilone & Longo, 2020; Marques-Silva & Ignatiev, 2022). As deployment into healthcare expands, greater interpretability of the factors that influence model predictions is essential to ensure safety and trustworthiness (Pawar et al., 2020; Srinivasu et al., 2022; Shaban-Nejad et al., 2021).

Medical imaging has seen a particular influx of deep learning applications in areas including: image classification, segmentation, synthesis, interpolation as well as denoising (Yousef et al., 2022; Aggarwal et al., 2021; Han, 2021; Haq, 2022). This is opposed to the rate of clinical adoption

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of these techniques, which is much smaller. A barrier to closing the gap between research and clinical adoption is the lack of interpretability (Martino & Delmastro, 2022). Consider a radiologist of the future who uses deep learning to help them assess magnetic resonance (MR) images. If the model predicts that a benign tumour will become malignant, but the radiologist strongly disagrees, without an interpretable explanation the output of the model is useless as there is no clear reason to trust its prediction.

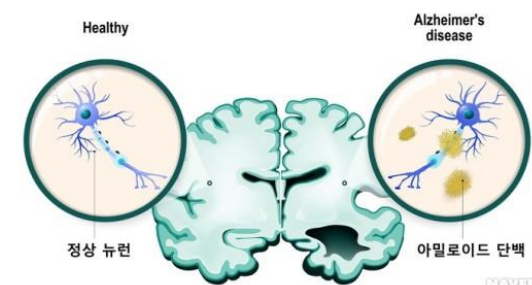
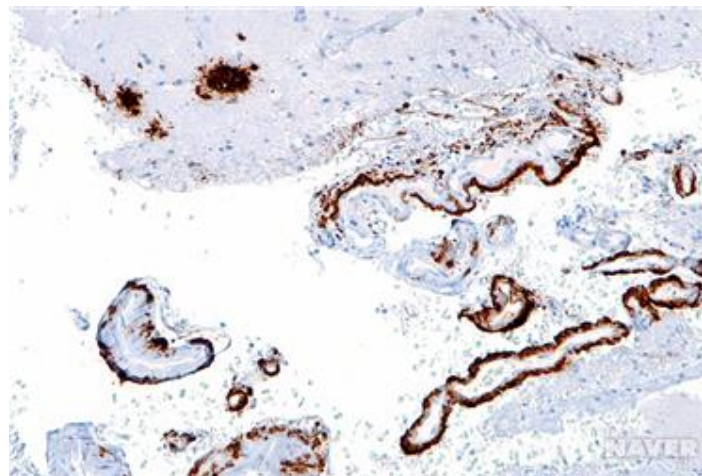
Beyond healthcare, concerns surrounding the interpretability of AI models have been raised by political bodies such as: the United Nations Educational, Scientific and Cultural Organization (UNESCO), the Organisation for Economic Co-operation and Development (OECD), the government of Australia as well as the United States and the European Union (Marques-Silva & Ignatiev, 2022; UNESCO, 2021; National Science and Technology Council (US), 2019; Gov, 2021; EU, 2021; OECD, 2021). As a response, a plethora of techniques in the field of Interpretable Machine Learning (IML) are being developed and the field can be discretised into two main approaches (Gautam et al., 2021). Post-hoc methods, which interpret the predictions of black-box models after they have made predictions, and self-explainable models (SEMs) whose predictions are accompanied by interpretable explanations.

The literature concerning the former approach is more extensive than the latter (Plumb et al., 2019; Rudin, 2018; Hedström et al., 2022). This is because SEMs have traditionally traded interpretability for accuracy, however recent methodological advancements have increased their predictive capabilities (Plumb et al., 2019; Gautam et al., 2022; Chen et al., 2018; Gautam et al., 2021). As such, we chose to develop an SEM in the present work, to contribute to this comparatively nascent field of IML.

One approach to producing computer vision SEMs is *prototype learning* (Rymarczyk et al., 2021; Kim et al., 2021; Nauta et al., 2020). Prototypes are in-class variants that provide transparency to model predictions because, the final prediction of the model is based on some similarity metric between input data and prototypes (Gautam et al., 2021; Chen et al., 2018). The intuition behind such models is that humans may classify visual objects based on how similar

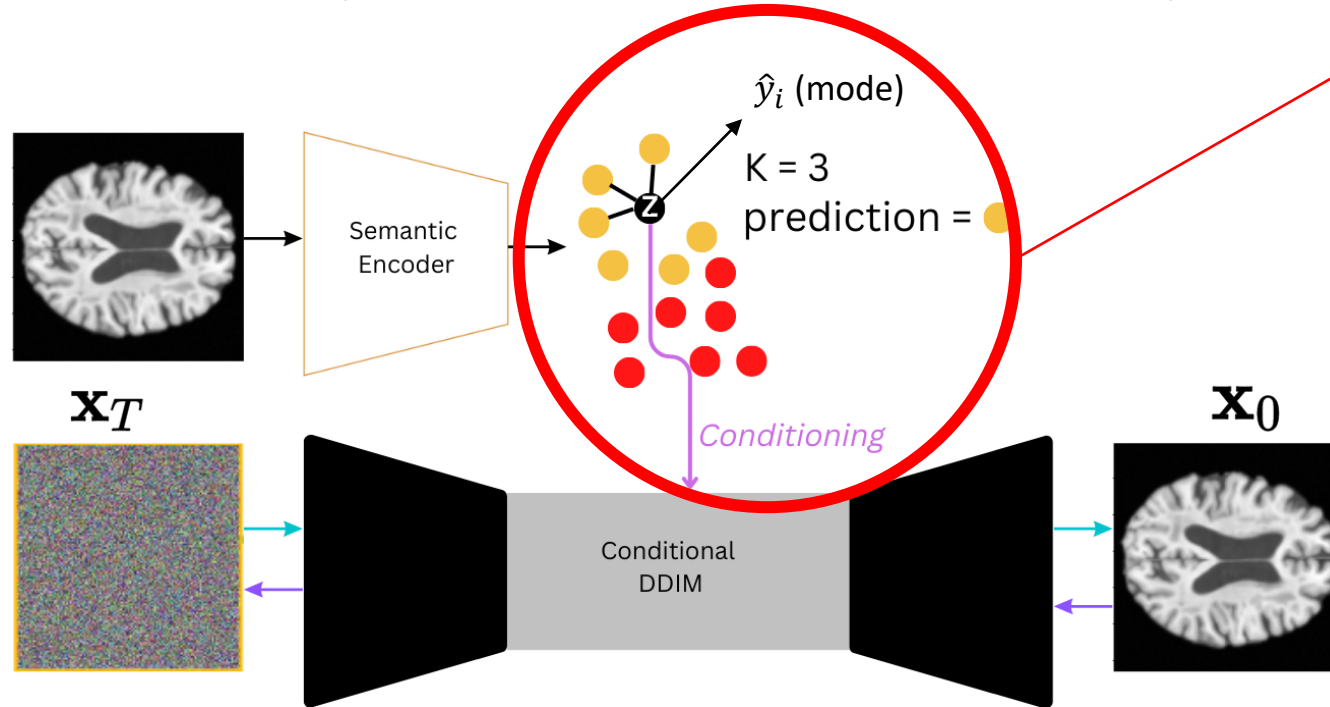
Alzheimer's disease(AD)란?

- 만성 신경 퇴행성 질환으로서, 아밀로이드반의 비정상적인 축적과 뇌의 신경섬유매듭이 엉키는 특징을 가지고 있음
- 대부분의 연구가 neuroimaging 데이터셋으로 이루어짐
- 현재까지 치료법이 발견되지 않았기 때문에 조기 발견이 매우 중요함
- 해당 연구에서는 **Alzheimer's disease(AD), Healthy Controls(HC)** 구분을 목적으로 함



4. Paper Review

3. Methods(DDIM model + Contrastive loss)



K-nearest method : contrastive learning

$$\mathcal{L}_{DIFF} = \sum_{t=1}^T \mathbb{E}_{\mathbf{x}_0, \epsilon_t} \left[\|\epsilon_{\theta}(\mathbf{x}_t, t, \mathbf{z}_{\text{sem}}) - \epsilon_t\|_2^2 \right]$$

-> DDIM model loss(MSE)

$$\mathcal{L}_{CONTRAST} = \frac{1}{B} \sum_{i=1}^B -\log \frac{e^{\text{sim}(\mathbf{z}^{p,i}, \mathbf{z}^{p,i})/\tau}}{\sum_{j=1}^M e^{\text{sim}(\mathbf{z}_j^{p,i}, \mathbf{z}_j^{n,i})/\tau}}$$

-> Contrastive loss

$$\mathcal{L}_{PRED} = \frac{1}{N} \sum_{i=1}^N \text{CE}(\hat{y}_i, y_i)$$

-> Predictive loss(Cross-entropy)

$$\mathcal{L}_{TOTAL} = \mathcal{L}_{DIFF} + \mathcal{L}_{CONTRAST} + \mathcal{L}_{PRED}$$

4. Paper Review

4. Experiments

DATA SET

- 10개의 공공 데이터셋(Ex ADNI) 을 사용하여 총 6137개의 환자 데이터를 확보함(AD = 1105, HC = 5032)

학습과정

1. Class 구분 없는 학습과정을 거침(340 epoch까지) – diffusion loss
2. K=7 latent representation learning을 활용(500 epoch까지) –Contrastive loss, predictive loss 추가

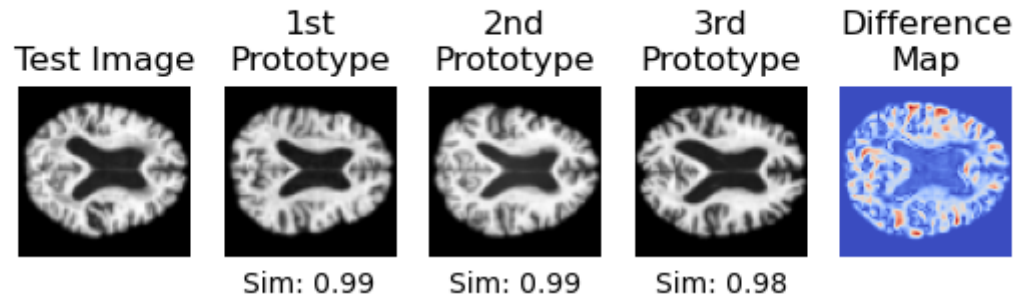
Hardware

- Preprocess model : U-Net
- Activation function : SiLU(Swish)
- Parameter number : 7.6 million
- GPU : Nvidia GeForce RTX 4090

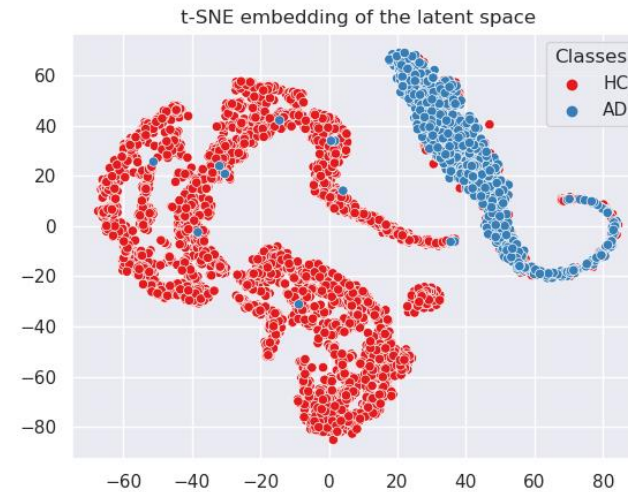
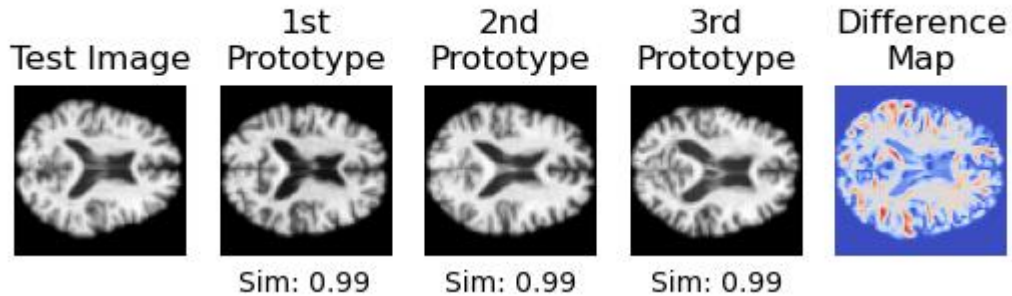
4. Paper Review

5. Results

Class: AD Predicted Class: AD



Class: HC Predicted Class: HC



MODEL	AUTHORS	ACCURACY	INTERPRETABLE
OUR MODEL	-	0.88	✓
CNN	HELALY ET AL. (2021)	0.97	×
SSAE	MENDOZA-LEON ET AL. (2019)	0.90	×
XCEPTION	TUFAIL ET AL. (2020)	0.81	×
CNN + SVM	SETHI ET AL. (2022)	0.88	×
SVM	UMA-RANI ET AL. (2021)	0.84	×
TCN	EBRAHIMI ET AL. (2021)	0.91	×

5. Conclusion

Results

- Diffusion이라는 생성모델 기법을 사용하여 모델의 해석력을 높임(Semantic representation)

Limitation

- Global explanation은 가능하나 local explanation은 불가능함
- 정확도 측면에서 기존의 baseline 모델보다 낮다는 한계점을 가짐
- Contrastive learning의 방법론 도입 여부에 대한 언급이 없음
- 임상적으로 해당 결과의 해석에 대한 검증이 이루어지지 않았음(image만으로는 판단이 어려웠음)

연구 활용방안

- 현재 연구중인 PPMI(파킨슨병) 데이터셋은 Multimodal dataset이고 특유의 라벨 적용이 어려움
- 따라서 해당 데이터셋에 diffusion autoencoder를 적용하여 latent variable을 추출할 예정
- 이를 이용하여 향후 RL(강화학습)에서 사용할 STATE를 Define할 예정

THANK YOU
